



Information systems control alignment: Complementary and conflicting systems development controls



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ABSTRACT

This study presents a new concept called information systems control alignment, which examines the degree that the underlying characteristics of four main information systems (IS) control dimensions are mutually complementary. Using three case studies, our research uncovers two high-functioning control patterns – one with traditional characteristics and one with agile characteristics – that demonstrate positive alignment among the control environment, control mechanisms, socio-emotional behaviors, and execution of controls. By better understanding the circumstances that contribute to control conflicts, organizations can be increasingly mindful of cultivating a complementary relationship among the control dimensions when designing, implementing, monitoring and adjusting controls within IS processes.

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1. Introduction

Organizations continue to struggle with ineffective information systems (IS) processes, as demonstrated by ongoing IS security breaches [8,32], underperforming IS development efforts [1,26], and inconsistent service from third-party providers [2,15,40]. In response, managers configure collections of IS controls that attempt to balance these concerns alongside efforts to achieve profitability and efficiency targets. However, the problem is that managers responsible for overseeing IS processes typically struggle to effectively design, implement, monitor, and adjust controls that are not performing as intended and not meeting organizational objectives [18,39,43,48]. We suggest that a possible explanation for declining IS process performance is a misalignment among four fundamental dimensions of control: the control environment (e.g., the structural, strategic and cultural factors that influence a choice of control), control mechanisms (e.g., the core characteristics of a policy or procedure), socio-emotional behaviors (e.g., the response that employees have to control mechanisms), and control execution (e.g., the effectiveness and subsequent adjustments that employees make to controls over time; [9,11–13,59]).

The purpose of this paper is to examine the new concept of IS control alignment, which refers to the degree to which the control environment, control mechanisms, socio-emotional behaviors, and control execution are mutually complementary within an IS process. In this context, we define a process as “a collection of practices influenced by the enterprise's policies and procedures that takes inputs from a number of sources, manipulates the inputs and produces outputs” ([27], p. 19). We suggest that conflict among any of these four control dimensions could result in staff frustration, unnecessary bureaucracy, and/or a lack of organizational agility. We pose the following research question: how do the underlying characteristics of IS control dimensions complement and/or conflict with one another? This exploratory study employs a qualitative approach using in-depth case studies of the IS development process at three organizations to investigate how controls are configured and to what extent elements of the control dimensions are found to align/complement or misalign/conflict with one another. We compare our results across the case studies to identify patterns that correspond with conflicting, as well as complementary, control dimensions. Based on these observations, we propose two optimal patterns of IS control that the empirical data suggest are highly complementary. One pattern, corresponding to traditional characteristics, advocates for the integration of a conventional control environment with preventive controls that leverage individual-oriented employee behaviors and a stable control structure. A second pattern, which corresponds to agile characteristics, finds a high degree of complementarity between a

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rapidly advancing control environment, detective control mechanisms, team-based employee behaviors, and an evolutionary control structure. Our findings suggest that IS processes following either one of these complementary patterns will generate positive performance benefits, whereas other patterns of control can lead to conflicts and declining performance.

Consider the example of an established financial institution with a well-run, traditional systems development process that remains relatively unchanged over time and employs experienced staff members with expertise in waterfall development methods. If company managers decide to abandon their current development approach for a set of procedures that adopts a highly iterative, agile philosophy, IS control alignment would suggest that these new control mechanisms would be at odds with the control environment (i.e., a bureaucratic, highly regulated organization), socio-emotional behaviors (i.e., staff have comfort and familiarity with the previous control mechanisms but not the new ones), and control execution (i.e., a systems infrastructure that has little need for rapid, ongoing adjustments). Refer to [Appendices A and B](#) for additional scenarios that illustrate organizational examples of strong and weak IS control alignment and control systems.

Existing research models provide important guidance in the selection of control mechanisms, including concepts such as a portfolio of control modes [34], control ambidexterity [58], and control balancing [23]. Although each of these concepts provides a valuable approach to addressing IS control problems, none simultaneously considers all four of the fundamental control dimensions: the compatibility of control mechanisms with the control environment, socio-emotional behaviors, and control execution. By introducing the concept of IS control alignment, the contribution of this study is to establish a holistic and inclusive approach that recognizes the inter-relatedness of IS control dimensions to address issues arising within IS processes, while remaining complementary to past models. This perspective provides a new lens that can contribute to more fully explaining why control mechanisms do not always operate as expected. For practitioners, the concept of IS control alignment can offer guidance when designing, implementing, monitoring, and adjusting controls. Rather than deciding on control mechanisms based on a limited set of factors (e.g., behavioral control should be used when IS project activities are easily observable), control alignment advocates for a tailored, more dynamic approach that enhances control performance by facilitating a complementary control environment, control mechanisms, socio-emotional behaviors, and control execution, while avoiding potential conflicts. In doing so, organizations may be increasingly able to avoid the control failures that have plagued systems development projects, as well as other IS processes, in recent years.

The remainder of this paper is organized as follows. The next section provides an overview of control concepts as well as alignment fundamentals. We then discuss in greater depth the concept of IS control alignment, including its key components. This discussion is followed by a presentation of our data collection and analysis methods. The findings are then presented, including case-specific details of the identified patterns of control alignment and misalignment. Finally, we propose two high-functioning patterns of control alignment, discuss our findings relative to past IS control research, and highlight a series of future research opportunities.

2. Theoretical and conceptual foundations

Our research integrates fundamentals from the study of both IS control and alignment in establishing the new concept of IS control alignment. We first review the concept of control and its theoretical underpinnings, then discuss alignment and introduce relevant past research that has been conducted in the field. These

concepts are then integrated, and an overview of IS control alignment is provided.

2.1. Control

Control is defined as *the attempt to affect the behavior of others as a means to achieve goals* [14,21,56]. Broadly applied, control is a topic of interest in a variety of managerial disciplines, including accounting, organizational behavior, strategy, and marketing [17,28,45,46].¹ This research concentrates on control within organizations, specifically on the execution and completion of information systems processes. In this context, IS control is defined as *the attempt to affect the behavior of others as a means to achieve objectives related to the development, operation, and use of information systems*. This viewpoint is inclusive of both managerial methods of controlling behavior (e.g., implementing an information security policy) and technical methods (e.g., software-based segregation of duties), which integrates the views of past research, including the studies by Orlikowski [44], Kirsch [34] and Tiwana [58].

Although originating in early studies, such as those of Davis [14], Fayol [19] and Urwick [60], the fundamental tenants of control are commonly attributed to Ouchi's [46] seminal work on the topic. Based on a framework for categorizing different types of control mechanisms (e.g., behavior, output, clan), the theory establishes a series of relationships between managers and the controls they choose to implement to influence employee behavior. Studies on control most commonly adopt one of two theoretical perspectives: an agency perspective or an organizational control perspective. The agency perspective focuses on the approach taken by a principal (the controller) to implement specific mechanisms that attempt to guide the divergent interests and actions of an agent (controlee; [58,17]). Within IS research, findings suggest that a variety of factors such as IS project stage, performance objectives, employee behavior measurability and employee knowledge drive the selection of particular controls [9,34,22,30,33]. Although this approach provides useful insights into selecting controls during particular situations (e.g., the type of controls that are best suited to the early stages of an outsourced systems development project), it is less useful in understanding the broader inter-relationships and implications of control choice and the organization, its processes, and its employees.

An alternative to the economics-based agency view of control is that of the organizational perspective, which places primary importance on the cybernetic process of managerial monitoring and evaluating of information to respond with control adjustments to achieve organizational goals [17,38]. In contrast to an agency-based view that establishes controls as a managerial mechanism to align principal and agent interests, the organizational perspective increasingly considers managers as the nerve center, with controls used to make adjustments to the ever-changing flow of information within the organization. Recent studies including those by Chua et al. [10], Gregory et al. [23] and Tiwana [58] consider issues that go beyond a traditional agency perspective by considering how controls work together and dynamically change over time. Such research establishes the importance of a more adaptive, dynamic view of control.

To establish a broad baseline for understanding the range of control factors to be considered by managers who are responsible for overseeing controls, we draw on an established control classification framework used in past research [11–13] that recognizes the key elements of both agency and organizational

¹ It is also recognized that the study of control has a long research history within the fields of philosophy, psychology, sociology, physical sciences, and military studies. The focus of this research, however, is on modern management disciplines.

Table 1
IS control dimensions (adapted from [11–13]).

Control dimensions	Control dimension description	Control dimension example
Control environment	The factors that determine why a particular control is chosen	A company's hierarchical structure is most commonly associated with controls that rely on management oversight and review
Control mechanisms	The defining characteristics of the controls	A new systems development methodology is implemented that is predominantly characterized by formal, preventive controls (e.g., documented policies)
Socio-emotional behaviors	The social and emotional response that employees have to particular controls	A new information systems policy that restricts the autonomy of employees and results in a decreased level of satisfaction is implemented
Control execution	The extent that a control is observed as operating effectively and the resulting control changes over time	IS controls are periodically evaluated by management, and adjustments are made to the control design where issues are identified

control theory. The framework consists of four dimensions of IS control phenomena in organizations: the control environment, control mechanisms, socio-emotional behaviors, and control execution (Table 1). The model was adopted in this study because of its broad-based view, where each dimension focuses on a unique component of IS control. Because IS control alignment is a new concept, an approach that has wide applicability is critical in establishing our research foundation.

The first dimension, control environment, refers to the organizational context associated with why a particular control is chosen. Factors such as organizational strategy [47], company structure [59], and process characteristics [34,33] have been shown in past research to play a role in determining the most effective type of control, as described below. For example, Kirsch [33] suggests that the observability of employee activities optimizes the use of behavior control, whereas the extent to which employee activities can be measured is related to the appropriateness of outcome control. This area of IS control research has uncovered an increasingly rich collection of factors that contribute to control choices, particularly in lieu of the changing nature of IS control in the context of outsourcing [40] and innovative approaches such as agile development [24].

Control mechanisms is the second dimension and refers to the defining characteristics of the IS controls themselves. Within the information systems literature, this dimension has been extensively examined with respect to two general categories of control, formal and informal, and four underlying control modes: behavioral, outcome, clan and self-control [9,34,10]. These groupings are commonly used as a classification system to differentiate between uses of control mechanisms in particular circumstances. For example, Rustagi et al. [50] examine formal controls in outsourcing partnerships, Chua et al. [10] consider clan controls in complex projects, and Tiwana [58] studies how formal and informal controls are complements or substitutes for one another. Recent studies, such as Gregory et al. [23], have worked to expand the traditional conceptualization of control mechanisms by further distinguishing between the characteristics of control types (procedural, social, hybrid), control degree (tight, relaxed), and control style (unilateral, bilateral).

The third dimension is socio-emotional behaviors, which is concerned with the socio-emotional response that employees have to particular types of controls. Although much of the IS control literature focuses on a controller choosing a control mechanism for particular circumstances that will derive organizational benefits, the role of the controlee in the process is often overlooked. Studies in this dimension consider how particular control characteristics are associated with the degree of employee satisfaction [52], employee motivation [53], and a worker's ability to be creative [20]. A subset of literature in this area considers the degree to which individuals do not comply with controls, which can have a significant negative impact on project or organizational performance [3,4,42].

The final dimension is control execution, which refers to the perceptions that a control is operating effectively and how these perceptions drive control changes over time. Past studies, such as those by Choudhury and Sabherwal [9] and Kirsch [35], establish the importance of adjusting controls during the course of systems development projects to address ineffective controls, changing project stages, and evolving controller–controlee relationships. In recent years, additional context has been uncovered regarding control execution, particularly with respect to the incremental decisions and actions taken by managers to respond to poorly performing controls, knowledge gaps, and inconsistent expectations [13,23]. Although ineffective controls (e.g., a security policy is implemented, but is not updated or enforced by management) are likely to correspond to broader process performance (e.g., overall quality or efficiency; [54]), the control execution dimension focuses on the identification of control inadequacies (e.g., a security policy is not preventing incidents due to a new form of online threat) and the resulting actions (e.g., periodic adjustments to the information security policy over time to account for new threats).

2.2. Alignment

The second conceptual element in our study is that of alignment, which refers to the “fit, match, agreement, or similarity between two conceptually distinct constructs” ([16], p. 51). Within the field of IS, the study of alignment has examined a group of unique constructs using a range of approaches. Drawing from Chan's [5] discussion of alignment within the IS discipline, three key categories of alignment research are identified: strategic, structural, and cultural. A fourth category, project alignment, is an emerging area that will also be discussed.

Strategic alignment refers to the degree to which business strategy and plans complement IS strategy and plans [6]. Strategic alignment is generally viewed as forming the most dominant subset of alignment research within the field and accounts for the most widely cited publications, including those by Chan et al. [7], Henderson and Venkatraman [25], and Reich and Benbasat [49]. The second category of IS alignment, structural alignment, refers to the degree of structural fit between IS and the business; the degree of structural fit is affected by the location of IS decision-making rights, reporting relationships, (de)centralization of IT, and the deployment of IS personnel [6]. The focus on particular structures when assessing this type of alignment varies within the literature. One view considers the differences between formal structures (e.g., mandated rules) and informal structures (e.g., employee connections) when examining structural alignment [5]. Another view focuses on the degree of centralization in IS decision making rights by a central IS group relative to decentralized groups [31]. The third category, cultural alignment, refers to the degree to which an organization or nation's culture is congruent with the organization's IS strategy and objectives. Work

in this area examines the links between company or national cultures and other organizational characteristics [5,37]. Finally, a more recently emerging type of alignment is project alignment, which refers to the degree to which an IS project's deliverables are congruent with the organization's IS strategy and project objectives [29]. Conceived from a process perspective, the achievement of IS project alignment is viewed as contributing to the execution of organizational strategy [29]. Our research builds on these approaches to alignment by framing the control alignment concept within the context of an IS process and with control constructs as the core elements being aligned. We explain this perspective in the following section.

2.3. Information systems control alignment

A new concept that integrates the theories of control and alignment is proposed in this study: IS control alignment. As Venkatraman [61] suggests, there are a variety of approaches to evaluating alignment between constructs, including matching, covariation, gestalts, and pattern/profile deviation.² We adopt a pattern deviation perspective, which suggests that multiple variables can be aligned with one another, where the degree of alignment can be determined by defining a set of “high-functioning patterns” either from theory or empirical data. Using this approach, IS control alignment represents the *degree to which the control environment, control mechanisms, socio-emotional behaviors, and control execution are mutually complementary within an IS process*. Our results, which are outlined in detail below, point to two high-functioning patterns – one with traditional characteristics and one with agile characteristics – that correspond to positive alignment among the four control dimensions. The identification of these IS control alignment patterns can aid organizations struggling with underperforming IS controls by providing guidance on adjustments that can be made to adopt control characteristics that complement one another.

IS control alignment is positioned in this research as a phenomenon that builds on past conceptualizations of IS alignment but is distinct in a number of fundamental ways. First, IS control alignment focuses on aligning constructs (i.e., IS control dimensions) that differ from past approaches, which investigate the alignment of strategies, organizational structure, culture, and project characteristics. By aligning these control constructs, IS control alignment suggests that improved outcomes will result. Second, IS control alignment has a distinct behavioral element that distinguishes it from past forms of alignment research. Rather than considering the broader organizational aspects of strategy or structure that drive the high-level direction of a company, control alignment takes a lower-level view of the approaches used to intentionally influence the behavior of employees during day-to-day processes. Finally, IS control alignment is considered distinct in the manner in which controls are viewed to rapidly change and evolve over time [9,34,35]. This type of alignment differs from other forms that include aspects of change but tend to occur more slowly over time (e.g., structural, cultural). In the following section, we further explain our approach to evaluating IS control alignment and provide a description of the data collection and analysis approach that we utilize. Two high-functioning patterns that emerge from our analysis are then presented.

3. Methodology

Our study employs a qualitative approach based on three in-depth case studies to identify characteristics of the four control dimensions, evaluate the extent to which the dimension characteristics align with one another, and determine the positive and/or negative outcomes that result from the control alignment/misalignment. Based on interviews and company documents, as well as the existing IS control literature, we iteratively develop a classification framework that distinguishes between the underlying characteristics of each of the four control dimensions. We then use the classifications as a foundation for identifying complementary and conflicting elements of control, which allows us to uncover the resulting positive and negative outcomes. Based on these findings, we establish a set of two patterns (consistent with Venkatraman's [61] pattern/profile deviation approach) that exemplify complementary control characteristics based on the empirical data. Our approach is discussed in greater detail below.

In this research, we set the unit of analysis at the process level. Rather than restricting our focus to a narrow subset of controls, such as within a single project, this approach allows us to consider a broader collection of controls across a range of related IS activities. In this context, examples of a process-level unit of analysis include managing security, managing enterprise architecture, and managing systems development. The IS process investigated in this research is managing systems development, which is defined as “establishing and maintaining solutions in line with enterprise requirements covering design, development, procurement/sourcing and partnering with suppliers and vendors” ([27], p. 132). This process includes activities such as systems configuration, test preparation, testing, and requirements management. The process was selected for investigation because it typically represents a large proportion of IS department spending and is also frequently associated with IS control issues that can be costly and risky to organizations [57]. Additionally, because much of the IS control literature focuses on the information systems development process, this area of focus can aid in extending past work in the field, as well as enable comparisons with past findings.

3.1. Classifying IS controls

The first step in evaluating IS control alignment is to establish the defining characteristics of each of the four control dimensions within the case study data. Case studies provide a rich and contextualized examination of organizational phenomena [62], which allows for a detailed assessment of IS controls. Because this study was exploratory in nature, the case sites were selected using a sampling strategy that sought to establish variation in each case's underlying control components [62]. To obtain such variation, participating organizations that varied with respect to four factors were sought: industry, company size, technical architecture, and systems development approach. The three participating organizations represent different industries (healthcare, insurance, and manufacturing), sizes (small, medium, large), architectures (mainframe, client-server, web-based), and development approaches (waterfall, hybrid, agile).

Case study data were collected using semi-structured interviews with employees involved in the systems development process as well as the review of organizational documents. A total of 20 interviews were conducted with senior management (e.g., VP Systems Development, Director of IT), middle management (e.g., Project Manager, Development Manager), and staff (e.g., Developer), consisting of over 13 h of discussion and comprising 412 pages of transcribed notes. Interviewees were selected based on their participation in the systems development process and were asked questions structured around the four control dimensions. For

² Venkatraman [61] uses the term “profile” to describe a coherent set of alignment characteristics. We believe that a term more appropriate to control alignment is “pattern” because it incorporates the notion of a profile and the inherent variability that is possible in control alignment characteristics. We use the term “pattern” in this paper.

Table 2
IS control classifications.

IS control dimension	IS control classification	IS control classification definition	Past use of similar control classifications
Control environment	Traditional	Traditional control environments are characterized by process standardization, established organizational structure, cautious decision making, and internal stability	Kling and Iacono [63]; McFarlan [64]; Rao et al. [47]
	Progressive	Progressive control environments are characterized by process flexibility, technology-intensive initiatives, risk-taking, and innovation	Kellogg et al. [65]; Sia and Neo [66]; Silva and Hirschheim [67]
Control mechanisms	Preventive	Preventive control mechanisms are proactive attempts to avoid the occurrence of an activity or incident that could potentially have a negative effect on the achievement of organizational objectives	Gopal and Sanders [68]; Phan et al. [69]; Kulp [70]; Nidumolu and Subramani [71]; Riehl [72]
	Detective	Detective control mechanisms are procedures conducted to identify whether an incident that could have a negative effect on organizational objectives has occurred	Gopal and Sanders [68]; Nidumolu and Subramani [71]
Socio-emotional behaviors	Individual	Individual socio-emotional behaviors refer to employees whose response to controls is consistent with a prioritization of individual goals, the focus on individual rights, and an emphasis on autonomy	Boss et al. [4]; Fitzgerald [20]; Love et al. [73]; Olsen [74]; Santana and Robey [52]; Morris and Marshall [75]
	Collective	Collective socio-emotional behaviors refer to employees whose response to controls is consistent with a priority on group goals and the cohesion of groups	Baronas and Louis [76]; Kirsch et al. [77]; Kraut and Streeter [78]; Orlikowski [44]; Stewart and Gosain [79]
Control execution	Stability	Stable control execution is characterized by controls that are viewed as performing adequately within mature, well-established processes. Few adjustments to controls are required, and controls are rarely added	Fitzgerald [20]; Kirsch [33]; Kirsch [34]; Kirsch et al. [80]
	Evolution	Evolutionary control execution is characterized by the identification of areas for control improvement that require remediation, as well as the associated adjustments and additions to controls over time	Choudhury and Sabherwal [9]; Kirsch [35]; Kohli and Kettinger [81]; Harris et al. [24]; Jin and Montague [82]

example, we asked participants about the types of control mechanisms used in the systems development process.

A total of 15 organizational documents (e.g., development specifications, development plans, test plans, application architecture) were also collected, comprising 123 pages. The use of multiple data collection methods allows for triangulation, that is, the verification of repeatability of an observation [55], which results in more convincing and accurate findings [62]. The documents were used as a data source that could corroborate the control-related data collected via interviews, as well as contribute additional refinements to the IS control classifications being developed.

Data coding was performed in three stages. First, we descriptively coded the interview transcripts and documents into categories corresponding to the four control dimensions noted above: control environment, control mechanisms, socio-emotional behaviors, and control execution. The objective of this exercise was to establish the existence of a four-dimension model of control within the empirical data. The coding process consisted of the first author tagging text segments of each interview transcript and document with one or more codes [41]. To verify the validity and reliability of the coding approach, an independent research assistant coded three interview transcripts and documents (i.e., one transcript and document from each case study) into the IS control dimension categories. The coding exercise resulted in an average inter-rater agreement Kappa of 0.73, which corresponds to a “substantial” strength of agreement range, as defined by Landis and Koch [36].

The second stage of the coding process focused on an assessment of the underlying characteristics of each IS control dimension through the use of pattern coding, which employs inference making as a means to explain a theme or configuration [62,41]. To better understand the defining elements of each dimension, two distinct lower-level classifications were established (i.e., eight classifications in total). Each classification represents a unique grouping of control characteristics that is

distinct from its “paired” classification and is established based on justification from the IS control literature.³ Table 2 outlines the underlying classifications used to analyze the collected data, including definitions and references from the literature that justify the groupings. These classifications were developed iteratively through data analysis, discussion between the authors, and references to similar classifications that exist in the control literature. Establishing these divergent classifications provided an opportunity to distinguish between the data identified in the control dimension coding and enhanced the ability to make cross-case comparisons. In total, 652 segments of text were coded during this exercise across all control classifications in both the interviews and documents.

The classifications used for the control environment dimension are traditional and progressive. Traditional control environments employ a high degree of process standardization (e.g., expectations that software will be developed to comply with a particular guideline), operate using a rigorous organizational structure (e.g., job titles and responsibilities are carefully defined) and have leadership stability. Decision making in traditional environments tends to be cautious, and managers are averse to accepting high levels of risk. In contrast, progressive environments encourage more risk-seeking, are more flexible in their process activities (e.g., openness to solving a problem in different ways), and are more accepting of innovation and technology-intensive initiatives.

Control mechanisms are divided into preventive and detective classifications. Preventive controls seek to limit the opportunity for issues to occur through means such as standard policies and procedures. Such an approach tends to work most effectively when

³ We acknowledge that the dimension characteristics typically range across a continuum (e.g., strongly traditional to moderately traditional), but for the sake of simplifying the analysis, only two categories for each dimension are utilized in this study. When a dimension included examples of both underlying classifications (e.g., traditional and progressive), we assigned the rating based on the degree of controls in each category, which considers the number, frequency and intensity of control characteristics [23].

managers are aware of what risks exist in a process and these risks remain relatively static. Control mechanisms can then be specifically designed to minimize these concerns. In contrast, detective controls are concerned less with the prevention of issues and focus instead on ensuring that issues that occur are identified and remediated through approaches such as management review. This approach is commonly adopted under circumstances in which there is rapid change within a process and new risks are regularly emerging. Rather than attempt to prevent an ever-changing list of concerns, detective controls remain open to a broad range of potential problems and address them as they occur.

Individual and collective classifications are defined for the socio-emotional behaviors dimension. Individual behaviors focus on employees who prioritize individual goals, rights, and autonomy. Under such circumstances, controls that allow employees to act independently are highly valued. For example, a manager may assign a task to a subordinate, but employees classified in the individual behavior category will prefer to execute the task independently. In comparison, collective socio-emotional behaviors refer to employees who respond positively to controls that are oriented toward group goals and cohesion. Employees in this group respond best to controls that motivate team interactions and enhance communication between team members (e.g., pair programming in agile development).

Finally, the control execution dimension consists of stable and evolutionary classifications. Stable control execution occurs when controls are perceived to perform in line with expectations and require few adjustments over time. For example, the data backup and recovery process in many companies is highly automated, reliable, and does not commonly require significant changes. In comparison, evolutionary control execution refers to IS processes that include controls that do not always work as intended and are frequently adjusted. In many companies, information security is a process that commonly identifies controls that are ineffective and require adjustments to stay current with frequently changing risks.

The final coding step examined evidence in the empirical data that suggested either a complement (i.e., alignment) or conflict (i.e., misalignment) between two of the classifications across any of the four dimensions. The objective of this exercise was to identify the classifications that appeared to be either complementary or conflicting, as well as to determine the positive or negative outcome of the alignment/misalignment. In line with our research question, we focused on interactions between dimensions, rather than within a single dimension. Complementary (aligned) controls represented a relationship between two or more of the control classifications that resulted in a *positive* organizational outcome. For example, if an interviewee commented that new systems development methodology procedures (e.g., a preventive control mechanism) had been broadly accepted by the systems development team (e.g., collective socio-emotional behaviors) and software product quality had been enhanced (i.e., the alignment outcome), we would code the passage as an example of complementary controls (i.e., preventive control mechanisms are complementary/aligned to collective socio-emotional behaviors). In contrast, conflicting (misaligned) controls were represented by a relationship between two or more of the control classifications that resulted in a *negative* outcome. For example, if an interviewee commented that the highly standardized nature of the organization (e.g., traditional control environment) was clashing with the implementation of team-based, monitoring controls (e.g., detective control mechanisms), which resulted in confusion and diminished efficiency (i.e., the misalignment outcome), we would code the passage as an example of a conflicting control (i.e., traditional control environments conflict/misalign with detective control mechanisms).

Our analysis focused on deriving insights from the data, and we remained open to evidence that suggested complementary or conflicting relationships among any of the classifications. Cases in which the classifications underlying the four control dimensions are complementary to one another are considered representative of control alignment, whereas control conflicts suggest control misalignment. A detailed cross-case analysis was conducted and discussed among the authors to identify patterns between the control classifications and evidence of control alignment/misalignment. In considering the theoretical relationships that these alignment/misalignment patterns were grounded in, we drew on the principles of Venkatraman's [61] approach, which seeks to establish the ideal circumstances in a scenario that will maximize performance of the process. By beginning with the 16 possible combinations of the eight control classifications (Table 2) across the four dimensions (Table 1), we followed Venkatraman's [61] approach in constructing alignment patterns from the empirical data. In total, 52 instances of either complementary or conflicting controls were noted, which we analyzed for evidence of positive alignment or negative misalignment. We present our findings in the following section.

4. Findings

In this section, we first present a brief overview of each case study setting. We then highlight representative instances from the empirical data of control classifications that complement one another, as well as those that conflict with one another. We then discuss the implications of these observations in Section 5.

4.1. The systems development process at HealthOrg

HealthOrg⁴ is a teaching hospital located in Canada. Approximately 80 IS staff operate using a model that employs a core interface system that links together nearly 60 otherwise unconnected clinical and business applications. A mix of mainframe, client-server, and web-based systems are used at the organization. Historically, HealthOrg has employed a traditional waterfall development approach but has been piloting a customized agile approach on a small number of selected projects. The team responsible for systems development consists of four main subgroups: a patient records team, a web development team, a dashboard/performance management team, and a database administration team. In addition, a corporate project management office (PMO) was recently added to the organization to oversee both IS and non-IS projects.

4.2. The systems development process at InsureCorp

InsureCorp is a medium-sized insurance and financial services company providing life insurance, investment services, and group benefits. InsureCorp employs approximately 750 staff members, with the IS department comprising approximately 120 staff members. The IS group focuses on both operational support (e.g., help desk, infrastructure maintenance, security) and IS project work. InsureCorp has traditionally employed a mainframe architecture but has also begun to adopt both client server and cloud-based architecture. The systems development process at InsureCorp was recently reorganized into four primary development competencies: business analysis, systems design, systems development, and quality assurance. The competencies are relatively siloed and sequential in nature, consistent with a traditional waterfall approach. After the reorganization, significant control structures were employed, which have since been scaled

⁴ Participant names have been disguised at the request of the organization.

back after they were determined to be too restrictive. Approximately 30–40 projects are ongoing at any one time at InsureCorp, 10 of which are considered large projects that are strategic in nature.

4.3. The systems development process at LargeMan

LargeMan is a manufacturing company based in the United States and employs approximately 75,000 employees worldwide. Systems development activities are decentralized, with a range of locations conducting tasks on particular systems used by the wider organization. In the Canadian subsidiary, where our data were collected, two key software applications related to business-to-business e-commerce transactions with distributors are developed and maintained by a local systems development team. Historically, LargeMan has employed a traditional waterfall systems development approach. However, the organization recently decided to experiment with a more iterative approach for some systems. Pilot projects have been conducted using an agile approach based on the Scrum methodology.

4.4. IS control alignment: Identifying complementary and conflicting IS control classifications

Our analysis focused on gaining insights from the empirical data that revealed both alignments (i.e., complementary control classifications across multiple dimensions resulting in positive outcomes) and misalignments (i.e., conflicting control classifications across multiple dimensions resulting in negative outcomes). As noted above, our analysis focused on the systems development process. The data presented below should be interpreted as an assessment of IS control alignment within a single process operating in three separate organizations. These results are presented in six groups⁵: (1 and 2) alignment and misalignment between control mechanisms and the control environment, (3 and 4) alignment and misalignment between control mechanisms and socio-emotional behaviors, and (5 and 6) alignment and misalignment between control mechanisms and control execution. Each group is explained in greater detail below, alongside representative interviewee quotes.

4.5. Positive outcomes from control mechanisms and control environment alignment

At LargeMan, multiple interviewees referred to situations in which the links between the control environment and control mechanisms contributed to organizational benefits. For example, interviewees noted two separate instances of business stakeholders requesting fast turnarounds on system modifications. In both cases, stakeholder expectations were exceeded relative to prior development approaches because of the increasingly agile-oriented control mechanisms in place. The development team was able to quickly respond to the requests and rapidly implement the necessary changes. In one instance, this response resulted in a significant financial benefit to the line of business. This situation demonstrates the alignment between a progressive control environment and detective control mechanisms. The following quotes describe the situations.

[One of] our [lines of business] ... needed something really quick because they were getting bazillions of calls into their call center to order these [products]. And they came to our team and normally it would have been, 'yeah okay, we will get that to you in 3 months' and I think it was 2 weeks or in 1 week they had a working demo. And then in another week or two they had it rolled into production and it was a \$1.5M return on business value. So that example ... actually [got] leaked into the [senior management] meeting, which is like my boss and all of the upper management for global and that actually came up in the middle of the meeting as an example. And from there it has just kind of worked its way through ... where the business goes, 'oh my gosh, you won't believe what the Canada team just did'. It is amazing, they were blown away. (Development Manager, LargeMan)

Essentially they said, 'we want this one piece of functionality for a China rollout' and we said, 'okay, but this is a big request and is going to take multiple sprints. Then 6 weeks later we called them up and said it is done because it took two sprints ... so yeah it is moving much faster than they are ready for so we end up with a lot of functionality that is in our back pocket that they have asked for but they are not ready to turn it on. (Development Lead, LargeMan)

HealthOrg interviewees also discussed an alignment between the control environment and control mechanism dimensions but one related to different control classifications other than those pertaining to LargeMan. As a hospital, HealthOrg was more strongly characterized as having a traditional environment that was more risk averse and structured. Because of the need for high reliability and accuracy in systems, this type of environment was viewed by interviewees as being most consistent with the use of preventive control mechanisms such as policies and procedures that actively anticipated and avoided negative issues from occurring. The following quote highlights this positive alignment between a traditional control environment and preventive control mechanisms.

On the clinical side of the house increasingly you find a lot of documented protocols. What does that mean? Well when we cut into somebody, before we do we ask them their name and we ask them which side they think that we are going to cut into. That may seem odd but in the banking sector you don't ask somebody whether they intend to take the money from this account. ... I mean maybe you do but you would not be asked five times before the money is actually extracted from the account. Here increasingly you are asked multiple times because there is recognition that systemic errors occur. That is it is not one point of failure but a combination of points of failure. So that is a classic example of a need for internal control, process controls. Because the processes are inherently risky. Once you cut you can't uncut. (Chief Privacy Officer, HealthOrg)

Evidence of complementary control mechanisms and control environment was also noted at InsureCorp. Interviewees highlighted the fit between the established, structured control environment and the ability to anticipate and plan for a finite collection of possible issues. By enacting control mechanisms to prevent these issues, the development process was considered to be more efficient than attempting to actively monitor and react to unexpected issues that are unlikely to occur. Similarly to HealthOrg, a link was suggested between a traditional control environment and preventive control mechanisms. The interviewee quote below highlights this perspective.

We do a lot of product projects and stuff like that. They are 20 thousand or 50 thousand dollars [and] ... typically they are very reproducible. If you do a re-pricing you know it is going to take

⁵ Although we view IS control alignment as the degree to which all four dimensions complement one another, in this section we present evidence supporting the alignment and misalignment between pairs of dimensions to more clearly focus on the key complementary and conflicting elements. However, we note that each pair of dimensions is also part of the broader inter-relationship between the remaining two dimensions, which can present additional opportunities for alignment or misalignment.

2 months and it is going to cost this amount of money and this what you know ... they can budget the whole thing based on historic estimates. So right when you get to this charter stage the project manager can pretty much tell you what it will look like. And in some cases we have not even bothered with a charter. We have just basically collapsed planning initiation down into one phase and just come up with a project management plan with the costs because basically a charter is just a high level of project management. So it depends on the size of the project. Now when I came into my role I realized that this is not formalized. It has basically taken a natural flow so one of the initiatives that I kicked off this year and we have just started this is the concept of right size. (Project Manager Director, InsureCorp)

4.6. Negative outcomes from control mechanisms and control environment misalignment

Despite instances of complementarity between control mechanisms and the environment at InsureCorp, there was also evidence of misalignment. For example, the systems development methodology, which had been undergoing a transition to a more iterative, agile approach, was noted by interviewees as being disconnected from the linear, structured project management approach. This finding suggests a partial departure from the preventive control mechanisms noted above toward the inclusion of increasingly detective control mechanisms. This misalignment with the control environment is noted in the following interviewee quote.

Because we don't know everything right from the outset ... there is a clash between the systems development lifecycle and what it possible for it versus the project management lifecycle, which by its nature is mandated to and thus wants to have target dates, deliverables, progress reports, and so on. And so there is sort of this natural tension that seems to happen between these two disciplines, especially as we undertake more and more projects. (Manager, InsureCorp)

Similarly, examples of control mechanism-environment misalignments were noted at HealthOrg. In one case, an interviewee highlighted the disconnect between the maturity of control mechanisms and the clinical focus of hospital management. In contrast to traditional organizations that have specialized managers leading organizational initiatives, HealthOrg had physicians working in senior leadership positions. Despite the high level of patient care that these individuals could provide, there was a degree of variability in their ability to advocate for the appropriate level of controls, which resulted in increased levels of organizational risk. The following quote highlights this perspective.

Presumably information management is helping and not hindering [the healthcare] process, not only on the administrative side but the clinical side [as well]. So it is a special case. It is not finance and it is not manufacturing. It is a completely different type of business and on the one hand then, you might think, well gosh, given the inherent riskiness of the sector and the fact that it has been going for so many years it must be extremely mature in its controls. Ahhhh, evidence is probably not so. At least there are varying levels demonstrably within the building, if not across the sector itself. One of the answers might be is because traditionally hospitals have been run by clinicians and they are not trained ... the same way administrators of banks are trained. The analogue might be that in banking who do you want running the bank? You want bankers running the bank. And in a hospital it is not so clear whether you want doctors running the hospital. (Executive, HealthOrg)

4.7. Positive outcomes from control mechanisms and socio-emotional behaviors alignment

Evidence of alignment between control mechanisms and socio-emotional behaviors was noted at LargeMan. Building on the adoption of more detective, agile methods of systems development, employees were able to become much more involved in the planning stages, which was viewed as contributing to a more dedicated, integrated team. By choosing control mechanisms that complemented the socio-emotional characteristics of employees, positive benefits were recognized, as described in the following interviewee quote.

Before [the adoption of agile methods] it used to be up to myself and maybe ... one other person on the team and you would come up with a high level estimate and that is what was given to our business people ... What I find with this process is we sit down and we figure out what we have and what we can do and everybody gets a say in how long they think that it is going to take and then everybody in the room sits down and you break that story down into individual tasks. So everybody gets to say ... in how long the task will take and at the end of the whole thing people feel that it is doable and we can get it done. I think that they are more focused and more committed. (Project Manager, LargeMan)

At InsureCorp, the importance of balancing the control mechanisms with socio-emotional behaviors was reinforced. In this case, interviewees recognized the importance of organizational culture in determining the appropriate control mechanisms. In cases in which a company has a low tolerance for risk, preventive control mechanisms are viewed as the best choice to best drive individual compliance toward organizational goals by minimizing the need to rely on individual employee trust. A key distinction between internal and external projects was noted, suggesting that differences were perceived to exist in the level of trust with vendors versus internal employees. The following interviewee quote discusses this link.

I think that there are cultural aspects and I think culturally if you are in an organization where risk is high and trust is low the default approach will be waterfall because I can get you to sign in blood and if we ever get into a problem ... when you are looking at external relationship, a requirements document it could be the basis of a court case in a worst case scenario, right? (Business Analysis Manager, InsureCorp)

4.8. Negative outcomes from control mechanisms and socio-emotional behaviors misalignment

Instances of control mechanism and socio-emotional behavior misalignments were also noted. At HealthOrg, a series of new controls (e.g., periodic reviews and remediations of identified risks) were met with strong resistance from staff. As a result, both executive and managerial inefficiencies were created to monitor and attempt to enforce the new controls. In this case, the detective controls were viewed as being misaligned with the individual autonomy (i.e., a socio-emotional behavior) of employees. The following interviewee quote explains the situation.

I thought that there would be and in some cases in the beginning there may have been some push back [to new controls]. 'Yeah, I hear you and thank you very much'. And if that happens, well when it comes to the CIO being through ... a risk report for this week and it is showing that we are still behind on the date that we said this would be completed. So he takes care of that too. The Directors don't want the CIO sending messages to themselves or to a team,

but that is the level of escalation that I need to take if they are not listening. (Risk Manager, HealthOrg)

Similar challenges were also noted at LargeMan. Although most staff were strongly in favor of the move to agile-oriented controls, a small group of senior developers with a background in traditional, waterfall approaches resisted the move. Concerns were raised regarding new approaches, such as pair programming and breaking down tasks into smaller activities, with which a few members of the development team were uncomfortable. For these team members, the adoption of new control mechanisms resulted in decreased satisfaction and increased stress. This view is discussed in the following quotes.

It is scary to think the thought of pair programming. Someone is going to be looking over my shoulder while I program, it is ... for some people it is daunting because whether or not they are self-conscious of their coding, they think that the person they are sitting with is better than them. But eventually you just have to get over those things and just go for it, but it is tough to tell someone who has been here 15 years, 'hey, let me look at your code'. (Developer, LargeMan)

The one item that I am trying to figure out how to address is that some of the more senior developers just feel like they are not satisfied. Just taking little tiny bits off the board ... they want something kind of meaty. (Development Manager, LargeMan)

4.9. Positive outcomes from control mechanisms and control execution alignment

Evidence of alignment between control mechanisms and control execution was noted at InsureCorp. Interviewees highlighted the desire for repeatability and consistency in executing their controls. Particularly with preventive controls such as policies and procedures, InsureCorp managers focused on employing operating stability. Ranging from steering committees to requirements documents, controls were standardized, mandatory for systems development initiatives, and remained stable and unchanged over time. The following interviewee quotes highlight this viewpoint.

You have a project steering committee and you want to have a systems development steering committee so that we can set the stage. 'Here is what we are expecting to see from you' and 'what are the opportunities for those improvements or you stop doing this because it doesn't make any sense and start doing this or change or continue this', so that we are actually heading in the right direction. (Vice President, InsureCorp)

When the group was formed I was appointed the manager of the group and one of the first things that we did was put together a basic requirements document template which has served us on most of the major projects that we have done since that time. (Manager, InsureCorp)

At LargeMan, a similar transition to align detective control mechanisms with evolutionary control execution had taken place as a result of the new agile methodology. Due to these new controls, a more reactive, dynamic approach could be taken to adapt to new information and circumstances, which was considered to incrementally improve project performance. In fact, project performance was perceived to increase over time because staff adapted to the new range of new information and perspectives that were able to be drawn upon. The following quote reinforces the importance of aligning control mechanisms with socio-emotional behaviors.

Early in the process there was hesitation to say that it was better, but I think over time people are realizing that we are not hitting those peaks where we are absolutely crunching to get things done.

But with that small amount of accountability comes a whole bunch more just by virtue of the fact that they now have a voice in the process. So [on] things like the sprint retrospectives, we get many people piping up and saying things who were very silent on the team before. It feels like they have more control over how we do things and how we can best utilize our skills to produce value. (Development Manager, LargeMan)

4.10. Negative outcomes from control mechanisms and control execution misalignment

At HealthOrg, evidence of a misalignment between control mechanisms and control execution was noted. Regarding the pilot evaluation of the new agile development approach, interviewees noted the challenges in balancing management's comfort with reliable, repetitive processes with increasingly adaptive and flexible controls. This view suggests a misalignment between stable control execution and detective control mechanisms because of a disconnect between established management expectations and how the controls actually operate. Such a misalignment can cause confusion and frustration, as noted in the following interviewee quote.

And of course there is always the struggle that management wants a plan, right? And they want to see progress and so it is difficult [for management] ... to understand that an agile process can change things along, but it takes time to get it right. And to have code that is bug free. And so it can be negatively perceived a little bit, as well as taking too long. Why is it taking too long? Why are there still bugs? Why is it still not working? And you almost want to hide it from them until we have it right. Or that we are not making change fast enough. (IT Manager, HealthOrg)

Evidence of control mechanism-execution misalignment was also noted at InsureCorp. Complaints arose from the business regarding how long it was taking to deliver IT projects, and the formal, document-intensive development methodology was viewed as a contributing factor. Because of the lack of flexibility in scaling the controls up or down for particular projects, InsureCorp's "one size fits all" approach was not complementary to the evolving needs of the business, which increasingly needed rapid delivery of technology solutions. The following quote describes the situation.

Right now what we are seeing the last 3 months or so we are getting a lot of pressure from our business partners because we have traditionally been working in the waterfall method and it is not good enough anymore. Things are taking way too long. Business requirements, 30 sessions, a 60-page document, which has what significance at the end of the game. (Vice President, InsureCorp)

This section has presented evidence from three case studies on the ways that the identified control classifications across four different control dimensions can complement or conflict with one another. In the next section, we discuss how these insights can be integrated into a model that can be used to guide organizations toward IS control alignment and how this approach can be positioned in light of recent research in the field.

5. Discussion

The results presented in the preceding section provide empirical insights into how the classifications underlying the four control dimensions complemented and conflicted with one another within the systems development process. Our findings suggest that although some of the control classifications are complementary to one another, conflicts exist with other

Table 3
IS control alignment patterns.

IS control alignment pattern	Control environment	Control mechanisms	Socio-emotional behaviors	Control execution
Traditional	Traditional	Preventive	Individual	Stable
Agile	Progressive	Detective	Collective	Evolutionary

classifications. Based on the results from our three case studies, we propose two high-functioning patterns (Table 3) that consolidate the complementary classification relationships that exist across the control dimensions. Although each of our case studies demonstrates different instances of control alignment and misalignment, these two patterns represent a baseline for maximizing the complementarity elements of the control dimensions, while minimizing the conflicting elements. That is, each pattern is supported by empirical data supporting positive relationships among all four classifications. This approach is in line with Venkatraman's [61] approach to alignment, which suggests that high-functioning patterns can be constructed based on empirical results. Table 4 summarizes the inter-dimension relationships that were identified in the Findings section by identifying the case studies and underlying classifications that correspond to both the traditional pattern (e.g., preventive mechanisms and traditional environment) and the agile pattern (e.g., detective mechanisms and progressive environment). Further support for the patterns is provided in Table 4 by identifying the case studies in which evidence of control classifications conflicts was identified (e.g., detective mechanisms and traditional environment). By consolidating these observations, the two patterns in Table 3 were established.

The traditional IS control alignment pattern integrates structured organizational factors (i.e., traditional control environment), effective anticipation and prevention of process issues (i.e., preventative control mechanisms), a hierarchical delivery of control in which individuals are tasked with responsibilities from supervisors (i.e., individual socio-emotional behaviors), and a relatively static collection of controls (i.e., stable control execution). In contrast to the focus on the structure and stability of the traditional pattern, the agile pattern instead assumes flexible and uncertain organizational factors (i.e., progressive control environment), an inability to predict control issues and thus a reliance on detection (i.e., detective control mechanisms), team-based process delivery in which collaboration and communication is key (i.e., collective socio-emotional behaviors), and a constantly changing collection of controls (i.e., evolutionary control execution).

Although we remained open to other possible combinations of control classifications that could result in additional patterns, our data indicated that conflicts existed between the remaining groupings, which could lead to sub-optimal performance (right-hand column in Table 4). However, because IS control alignment

considers the *degree* to which control dimensions complement one another, we are careful to note the possible range of misalignments relative to the two high-functioning patterns. Pattern deviation suggests that the most benefits will accrue for a process that complies exactly with the pattern configuration but also that performance will become relatively stronger the closer a process is to complying with a high-functioning pattern, even if it does not fully meet all classifications [61]. Therefore, alignment theory suggests that a systems development process characterized by a traditional environment (traditional pattern), preventive mechanisms (traditional pattern), individual socio-emotional behaviors (traditional pattern), and evolutionary execution (agile pattern) (i.e., three traditional classifications, one agile) will perform better than a process characterized by a traditional environment (traditional pattern), preventive mechanisms (traditional pattern), collective socio-emotional behaviors (agile pattern), and evolutionary execution (agile pattern; i.e., two traditional classifications, two agile).

These two patterns can act as a guide for organizations seeking to reduce the conflicts between control dimensions within their IS processes in circumstances in which controls are being designed, implemented, monitored or adjusted. Because past research recognizes the dynamic nature of IS controls, the patterns do not suggest that IS controls should remain static but rather that the overarching characteristics (i.e., classifications) of controls within each dimension will be increasingly complementary with other dimensions when they follow one of the patterns.

The IS control alignment concept that we present in this study is one of a variety of perspectives that can aid in understanding how IS controls are designed, implemented, monitored, and adjusted in organizations. We suggest that our results provide a supplement to other perspectives, such as portfolios of control [34], control ambidexterity [58], and control balancing [23], by providing a unique lens through which to view IS control phenomena (see Table 5).

A portfolio of control modes [34] is a popular perspective in the research literature that draws on Ouchi's [46] conceptualization of control into formal (behavior and outcome modes) and informal (clan and self-modes) groupings. Control portfolios are collections of control mechanisms that are selected based on factors such as task characteristics, role expectations, and project-related knowledge. IS control alignment is consistent with the control portfolio concept in that it positions the key factors (task characteristics,

Table 4
IS control alignment patterns.

Control dimension relationships	Support for traditional pattern classifications	Support for agile pattern classifications	Support for non-pattern conflict classifications
Control mechanisms–Control environment	HealthOrg (preventive mechanisms + traditional environment)	LargeMan (detective mechanisms + progressive environment)	InsureCorp (detective mechanisms + traditional environment)
Control mechanisms–Socio-emotional behaviors	InsureCorp (preventive mechanisms + individual socio-emotional behaviors)	LargeMan (detective mechanisms + collective socio-emotional behaviors)	HealthOrg (detective mechanisms + individual socio-emotional behaviors)
Control mechanisms–Control execution	InsureCorp (preventive mechanisms + stable execution)	LargeMan (detective mechanisms + evolutionary execution)	
HealthOrg (detective mechanisms + stable execution)			

Table 5
IS control perspective comparison.

IS control perspective	Summary of IS control perspective	Comparison to IS control alignment
Portfolio of control modes [34]	Based on task characteristics, role expectations, and knowledge, formal or informal control mechanisms are selected to form an integrated portfolio	Control portfolios recognize the close relationship between the control environment (e.g., task characteristics) and control mechanisms but does not directly consider the importance of socio-emotional behaviors or control execution
Control ambidexterity [58]	Considers the extent to which formal and informal controls act as complements or substitutes for one another	Whereas control ambidexterity essentially examines how control mechanisms align with other control mechanisms, IS control alignment considers how control mechanisms align with the control environment, socio-emotional behaviors, and control execution
Control balancing [23]	Control mechanisms are configured based on the categorizations of control type (procedural or social), control degree (tight or relaxed), and control style (unilateral or bilateral), which are incrementally adjusted through a series of sequential phases	Both control balancing and control alignment consider the link between control mechanisms and control execution; however, control alignment also examines links to the control environment and socio-emotional behaviors

etc.) as the control environment dimension, whose aims is to align with the control mechanisms dimension. IS control alignment differs in that it also considers the degree of alignment with socio-emotional behaviors and control execution. For example, consistent with a control portfolio approach, LargeMan adopted an agile development control mechanisms due in part to control environment factors such as the need to respond quickly to customer requests. However, IS control alignment highlights that these new control mechanisms also had a positive effect on employees (i.e., socio-emotional behaviors) and were in a state of ongoing adaptation since implementation (i.e., control execution). That is, control alignment adopts a broader, more holistic view that considers how the act of choosing controls can be aligned (i.e., alignment between control environment and control mechanisms) but also how compatible controls are with employee well-being (i.e., alignment between socio-emotional behaviors and control mechanisms) and changes over time (i.e., alignment between control execution and control mechanisms).

Control ambidexterity provides another perspective that considers the extent to which control mechanisms are complements to or substitutes for other control mechanisms [58]. Building on Kirsch's [34] control portfolio concept, control ambidexterity examines formal and informal control interactions within the same portfolio. Findings suggest that informal control mechanisms complement formal behavior control mechanisms but act as a substitute for formal outcome control mechanisms. For example, control ambidexterity could provide useful insights into the examination of control mechanisms at LargeMan, where behavior controls and outcome controls have recently given way to an increasing number of informal controls (e.g., clan-based agile methods, such as self-determining teams). In contrast to alignment considerations *within* the control mechanisms dimension alone, we consider IS control alignment representative of the complementary relationship *among* the control environment, control mechanisms, socio-emotional behaviors, and control execution dimensions. In the context of LargeMan, this view includes how the use of agile control mechanisms complemented the organizational structure, employee characteristics, and ongoing organizational change. In this respect, control alignment and control ambidexterity are not inconsistent with one another; rather, they examine control compatibility at different levels.

A third perspective is control balancing, which suggests that control mechanisms are configured based on the categorizations of control type (procedural or social), control degree (tight or relaxed), and control style (unilateral or bilateral). Drawing on past views of the modification of controls over time, such as the view of Choudhury and Sabherwal [9], control balancing argues that a series of sequential phases that provide an opportunity for incremental adjustments of control type, degree, and style is

triggered. At LargeMan, control balancing could provide insights into the steps taken during the transition from waterfall-oriented control mechanisms to agile-oriented control mechanisms. Both control balancing and control alignment consider the link between control mechanisms and control execution; however, control alignment also considers links to the control environment and socio-emotional behaviors (e.g., complements between LargeMan's agile control mechanisms and the organizational structure and employee characteristics). Similarly to the comparison to control portfolios, the control alignment concept offers a more holistic, multi-dimensional view than control balancing.

In general, we perceive IS control alignment as providing a complementary perspective to control portfolios, control ambidexterity, and control balancing. However, control alignment is distinct in its higher-level, holistic view of all four control dimensions, similarly to control portfolios and control balancing, which consider the relationships between two dimensions and control ambidexterity that examines a single dimension. Although each of these perspectives provides valuable insights, control alignment contributes a unique view by considering the degree to which the four dimensions are simultaneously compatible with one another. From a research standpoint, this framework provides a new way of considering the complex relationships between control phenomena. This multi-dimensional view provides a means to more fully explain why control mechanisms can range from highly effective to decidedly weak. For practitioners, the concept of IS control alignment and the classification framework (Table 2) can provide a guideline to consider when designing, implementing, monitoring and adjusting controls. Rather than deciding on controls based on relationships with a limited set of factors, control alignment suggests that organizations should more fully consider a holistic approach that encourages the control environment, control mechanisms, socio-emotional behaviors, and control execution to be mutually complementary.

6. Conclusion

This research was motivated by the ongoing challenges experienced by organizations in controlling their information systems processes and the opportunity to improve how controls are designed, implemented, monitored, and adjusted. Issues of security breakdowns, outsourcing disputes, and failing development projects demonstrate the failure of IS controls to aid in the achievement of objectives for organizations. Despite controls being recognized as an important driver of process performance, little guidance is currently provided to managers to establish a holistic approach that encourages different dimensions of IS control to complement one another. Motivated by this practical problem, this exploratory study draws on control and alignment theory to

investigate how different dimensions of IS controls complement or conflict with one another. Our data-driven results suggest that instances of both control alignment and misalignment are widespread. We propose two high-functioning patterns of control that consolidate a unique collection of control classifications that our empirical findings suggest are mutually complementary. The control alignment concept extends past perspectives of IS control by advocating for a holistic, multi-dimensional integration of control elements.

As with any research initiative, this work is subject to limitations. First, we recognize that control classifications other than the ones we employed could be used to distinguish between the control dimensions. Although the classifications we use are well established in the literature (Table 2), employing additional classifications, such as a tight versus relaxed control degree [23], could generate a more complex, multi-dimensional view of alignment, which could also contribute to the ongoing theoretical development of the control alignment concept. Second, although we attempted to select case study organizations that represented a range of industries, sizes, and systems development approaches, it is unclear how the IS control alignment concept may translate to other types of organizations. It is possible that particular industries or companies within highly regulated environments or highly volatile environments may already operate at an extreme level of traditional or agile control alignment. Such companies could have little opportunity for improvement or for differentiating themselves from competitors who also operate at a highly effective degree of alignment. Finally, the cross-sectional perspective of control alignment that our study presents would be enhanced by adopting a more longitudinal view. Consistent with past views on strategic alignment [6,51], the exploratory view of control alignment that we adopt in this study is consistent with the concept of an ongoing, dynamic phenomenon. By tracking how alignment evolves over time, further insights can be gained into how this view can be evaluated and improved in organizations.

Several research paths stem from this study. First, because of the ongoing innovations and growing importance of IS processes outside of systems development, a more comprehensive examination of controls within areas such as information security management or enterprise architecture may be desirable. By applying IS control alignment concepts to a wider range of IS processes, researchers may uncover unique aspects of IS controls that do not exist in a development context, and practitioners can improve controls in other important areas of their organizations. Future research could also seek to refine the approach used in this study by further considering additional patterns to the two we propose. Considering the addition of supplementary control classifications could also aid in developing the specificity of the elements that complement or conflict with one another. Finally, although this research outlines the first steps in the development of the IS control alignment concept, an even more tangible and scalable tool that could be used by practitioners to measure and respond to IS control alignment issues would be of significant value. Such a tool could take the form of a survey for collecting information on the control dimensions and classifications within a particular IS process, which could then aid managers in identifying areas that represent control misalignments and provide useful guidance in making adjustments that could bring controls back into alignment.

The objective of this study was to consider how the new concept of IS control alignment could aid in uncovering control dimensions that complement and/or conflict with one another. Using a case study approach at three organizations, we identify the core elements of IS control within the systems development process and propose two distinct patterns of IS control alignment. Our findings suggest that control dimensions will complement each

other when they are grouped by classifications of either agile or traditional attributes. By adopting a more mindful approach to developing collections of controls that complement one another across all four dimensions, organizations can position their IS processes to avoid the control failures that have diminished past levels of IS process performance.

Appendix A. Strong IS control alignment scenario

Strong IS control alignment scenario

AccountAssure is a mid-sized accounting company that provides audit, tax, and consulting services to a variety of local organizations. The company is characterized by a very traditional, hierarchical structure that abides by very rigorous, documented processes. Last year, AccountAssure initiated a project to implement a new application to be used by staff conducting accounting engagements with clients. The new application, to be developed internally, would replace another application created by the company 10 years ago. The project was carefully planned out, with the necessary budgets, timelines, staffing, and development approach documented and approved by AccountAssure partners prior to commencement. Using careful managerial oversight, the project was monitored using a series of checkpoints throughout its duration. IS staff were familiar with the process because the development approach had remained unchanged since the last system was developed. The staff felt that the process produced good-quality results on a timely basis and were reluctant to try out any of the new, iterative approaches to development that had emerged at other companies. The new software was delivered within the planned timeline and budget constraints. In a post-implementation review, the Internal Audit group attributed much of the project's success to the similarities between the highly structured environment, the nature of the controls in place, the employee expectations, and the stable collection of controls

Appendix B. Weak IS control alignment scenario

Weak IS control alignment scenario

Call-All is a mid-sized company that provides outsourced call center services to a range of clients. The company is characterized by a very traditional, hierarchical structure that abides by very rigorous, documented processes. Last year, Call-All initiated a project to implement a new web-based information system that would allow the company to manage client data more easily. Much of the business that the company had previously done on an older, legacy system would transition to this new platform. The organization's IS staff did not have much experience with web-based development, and there was uncertainty about the new system's requirements, which seemed to change regularly. In the tradition of the company, highly structured, rigorous procedures were put in place to design and test the system. Months went by, and the project slipped further and further behind schedule as new features were introduced and the process became bogged down in bureaucracy. The IS staff became frustrated that their development approach was not able to adapt to the rapid changes within the project and complained to management that an iterative approach may be more suitable. After investing 12 months and nearly one million dollars, the project was discontinued, and the old system remained in place. The Internal Audit department was asked to perform an evaluation of the project and found that the disconnect between the highly structured environment, the nature of the controls in place, the employee expectations, and the project's pace of change were responsible for the failure of the initiative

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